



Hope Kim @rlaope

Software Engineer

In 2014, a kid who dreamed of being a scientist in a white lab coat discovered computers. I don't believe in limits that haven't been mathematically proven. The harder the problem, the more exciting.

At 17, I took on a government-funded project – unfamiliar domains, tight deadlines and resource constraints, everyone thought it was impossible. Under pressure, I dove deeper into system development, grew alongside my teammates, secured \$33K investment, and built a new project room at Software Meister High School.

Currently at HonestAI as a server engineer – building savings bank online linked investment systems, integrating external financial APIs (NICE/KCB, Shinhan, KFTC, Toss, KakaoPay), JVM/K8s observability infrastructure, TCP tuning on financial messaging servers, and load testing scaling from 1K to 500K transactions.

HISTORY

2014. Junior

First software development in elementary school. Built 'FLY', an RPG game.

Software Meister High School (CSE). 2022 ~ 2024

Researched and validated server systems, JVM internals, and open-source systems.

10+ Projects.

School OAuth, nationwide dormitory management, hackathon AWS award, SW-Wave booth.

Government Project. 2023. 09 ~ 2024. 11

Vocational education platform for specialized high schools. Led server development with 80+ partner companies and 16+ affiliated agencies. Secured 45M KRW investment.

HonestAI. 2024. Dec ~ Now

Server Developer at Vertical AI fintech. B2B BaaS linked investment system, fault detection rate improved 40% to 90%, G1GC migration reducing P99 from 2~3s to 0.4s, build time optimization from 20m to 1~3m.

OPEN SOURCE

Argus JFR-streaming profiler for JVM 21+. Virtual Threads monitoring, lock-free ring buffer, WebSocket real-time dashboard for production JVM analysis.	kar98k Go traffic simulator for chaos engineering. Supports HTTP/1.1, HTTP/2, gRPC protocols. Poisson distribution spike generation, TUI interface with P50/P95/P99 latency tracking.	claude-code-savant Claude Code plugin system with 5 genius personas (Einstein, Shakespeare, Socrates, Steve Jobs), 16 dev specialist agents, 14 business agents. Includes web chat UI.
airML Lightweight ML inference runtime for ONNX models. ~50MB binary size, 0.01s cold start latency. Apple Silicon native optimization with Metal acceleration.	Mascot IntelliJ IDEA mascot plugin published on JetBrains Marketplace. Adds customizable character overlays to the IDE workspace.	cache-labs Cache architecture design study. Local + Redis dual-layer caching with Circuit Breaker pattern. Includes high-traffic scenario benchmarks and failover tests.
g1-log-capture G1GC log analyzer tool. Detects STW pause anomalies, provides GC tuning suggestions, and generates visual timeline output for GC behavior analysis.	tcp-css-loadtest TCP mock server for financial system load testing. Implements Nice protocol specification with EUC-KR encoding for Korean financial messaging.	TatVT0N Diffusion-based tattoo virtual try-on system. SAM 2 + SDXL + ControlNet + IP-Adapter pipeline. Optimized to run on consumer GPUs with 12GB VRAM.

PRESENTATION

KotlinConf'24 Global. 2024

JetBrains KotlinConf Global in South Korea. Presented on Project Valhalla, JVM Type System evolution, and Value Class performance benchmarks.

IT Union Conference. 2025

Software Meister High School IT Union conference. Carpe Diem – on cultivating software development skills while preserving individuality.

GSM DevConf. 2023

Growth of Developer – exploring the developer's growth curve, choosing direction, and the mindset needed to pursue long-term mastery.

INTERESTED

JVMGC

SRE

AI (ML/DL)

Cloud

Network

OS

Art

The act of satisfying someone through engineering

CONTACT

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HonestAI

Software Engineer – server/infra engineering, 2024. Dec ~ 2026. Mar

A detailed document describing the problems, considerations, solutions, and results encountered while working at HonestAI.

Responsibilities are in the Resume below; detailed write-ups are narrowed down to three topics.

Resume

SERVER

- Developed/operated savings bank online linked investment system
- Integrated external APIs – Savings Bank Central, NICE/KCB, Shinhan Bank, KFTC, Toss, KakaoPay
- Loan system traffic **1,000 → 500K** load testing, infra scaling, resolved Netty bottleneck (jstack)
- Built linked investment fault alert system – **detection rate 40% → 90%**
- Investment inquiry → confirm → deposit → execution fault response – **Read-Time 15m → 30s**
- NICE credit batch data **100MB → 1GB** expansion – bandwidth saturation response, traffic control
- TCP tuning on financial messaging server – socket pool reuse, Nagle disable, sub-ms latency gains

INFRA

- JVM/K8s Monitoring & Observability – Prometheus, Grafana, Micrometer
- Log Observability – Loki + Grafana dashboards
- Athena Partition Projection redesign – **log query 4~5m → 1~4s**
- Legacy JVM Observability (jmx-prometheus) & tuning (jcmd, jstack, jmap)
- Settlement batch **Full GC Hang** detection, memory tuning, alert manager system
- G1GC migration – **P99 2~3s → 0.4s**, OpenJDK 8 optimization benchmarks
- Gradle parallelism/cache + daemon heap tuning – **build 20m → 1~3m**
- Micrometer domain error aggregation – fault frequency & priority analysis
- Core banking DB lock timewait, deadlock monitoring & improvement

FINANCE

- Synced HonestFund core banking bond ledger with savings bank linked investment ledger
- Managed repayment tasks, special realization & acceleration bond schedules
- Credit bureau, NICE/KCB, KFTC bond status sync & concurrent loan registration
- Loan application-to-disbursement system & **auto-loan via R-Clips credit scoring**

For detailed write-ups → rlaope.github.io/honest.html

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Government Project

Backend Lead – Vocational Education Platform for Specialized High Schools, 2023. 09 ~ 2024. 11

At 17, I jumped into a government-funded project. Built a vocational education & credit management platform (School-of-Company) for specialized high school students. Led server development and collaborated with affiliated organizations.

Resume

SERVER

- Developed/led vocational education & credit management server system
- JPA Persistable, hashCode>equals override – **N+1 query optimization** (prevented extra queries on create)
- Led server dev and technology consensus from initial design to production
- Wrote Architecture Decision Records, facilitated stakeholder consensus
- 80+ partner companies, 16+ affiliated agencies** – developed features reflecting VoC

QA

- End-to-end integration testing: signup → course enrollment → credit management
- QA data set (15x/day, ~5min each) automation → **~5 second simulation**

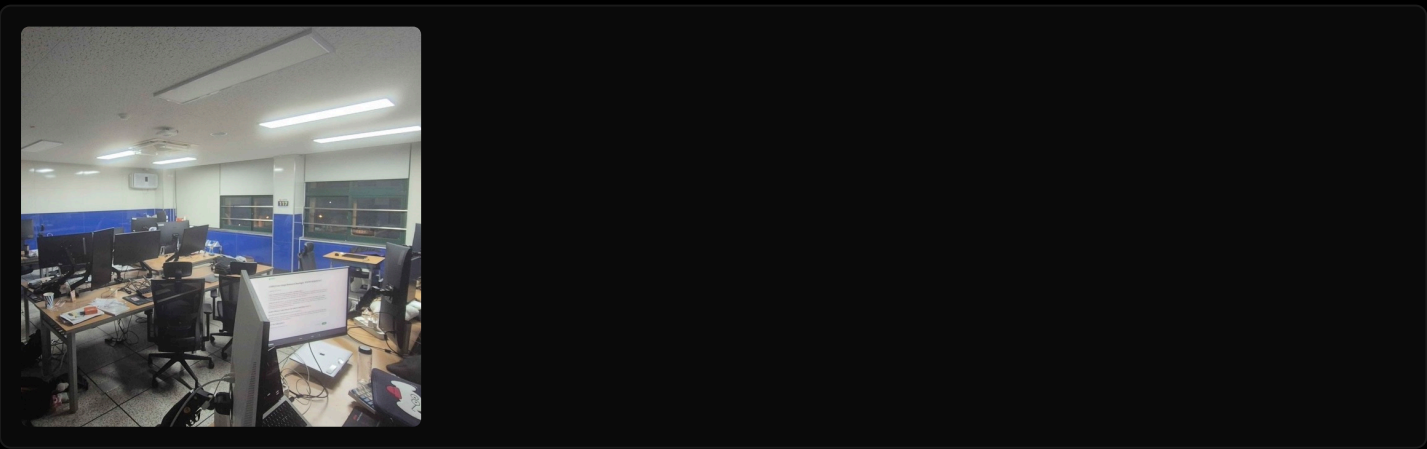
INFRA & PRODUCTIVITY

- PR priority system + Discord Bot Reminder – resolved PR neglect
- Full server infra design, **Terraform IaC**, deployment/ops then handoff to team

Achievements

- + Government-funded – Board of Education specialized high school credit certification platform
- + After successful delivery, **secured \$33K (45M KRW) investment**
- + Built a new project room at Software Meister High School
- + Handed off operations to the next cohort

School Company Room



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